BASE CHANGE OF THE NAIVE COTANGENT COMPLEX IS A QUASI-ISOMORPHISM OVER A FLAT ALGEBRA, AND DERIVED BASE CHANGE WORKS UNDER A TOR AMPLITUDE BOUND

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 22809-22823 (canonical tag 0FJU) and the counted proof is the proof environment at source lines 22825-22853, which immediately follows it and is its own proof.

Editorial note. The statement displays the cocartesian square of rings in the author's own Xy-pic markup, which is compiled from that markup with the shared preamble, and the naive cotangent complex macro is supplied by the retained preamble.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

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Allowed standard prerequisites (editorial).

- A presentation of the algebra by a polynomial algebra over the base gives a two-term complex computing the naive cotangent complex, with the conormal module in degree -1. used at proof lines 22825-22827 and 22845-22851. Fixes the explicit complex whose base change is compared, and expresses the tor amplitude condition through the flatness of the conormal module.
- For a flat algebra the kernel of the base changed presentation is the tensor product of the original kernel with the new base ring. used at proof lines 22828-22833. Identifies the conormal module of the base changed presentation with the base change of the original one, which gives the isomorphism of complexes.
- A two-term complex whose degree zero term is free has tor amplitude in the interval from minus one to zero exactly when its degree minus one term is flat, so the derived tensor product coincides with the ordinary one. invoked at proof lines 22849-22852. Turns the ordinary base change isomorphism into a derived one under the tor amplitude hypothesis.

Source reference. Original section 00S0: sources/stacks-chow/tags.txt, source lines 1022–1022. The reference points to the place where presentations exhibiting the

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naive cotangent complex are set up, cited at proof lines 22825-22827; it is a pointer to the framework and the computation is inside the counted span.

Source reference. Original lemma 0653: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16883–16883. The reference converts the tor amplitude hypothesis into flatness of the conormal module, used at proof lines 22849-22850; the conclusion drawn from that flatness is inside the counted span.

Lemma 0.2. *Consider a cocartesian diagram of rings*

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array}$$

If B is flat over A, then the canonical map $NL_{B/A} \otimes_B B' \to NL_{B'/A'}$ is a quasi-isomorphism. If in addition $NL_{B/A}$ has tor-amplitude in $[-1, 0]$ then $NL_{B/A} \otimes^{\mathbf{L}}_B B' \to NL_{B'/A'}$ is a quasi-isomorphism too.

Proof. Choose a presentation $\alpha : P \to B$ as in Algebra, Section 00S0. Let $I = \mathrm{Ker}(\alpha)$. Set $P' = P \otimes_A A'$ and denote $\alpha' : P' \to B'$ the corresponding presentation of B' over A'. As B is flat over A we see that $I' = \mathrm{Ker}(\alpha')$ is equal to $I \otimes_A A'$. Hence

$$I'/(I')^2 = \mathrm{Coker}(I^2 \otimes_A A' \to I \otimes_A A') = I/I^2 \otimes_A A' = I/I^2 \otimes_B B'$$

We have $\Omega_{P'/A'} = \Omega_{P/A} \otimes_A A'$ because both sides have the same basis. It follows that $\Omega_{P'/A'} \otimes_{P'} B' = \Omega_{P/A} \otimes_P B \otimes_B B'$. This proves that $NL(\alpha) \otimes_B B' \to NL(\alpha')$ is an isomorphism of complexes and hence the first statement holds.

We have

$$NL(\alpha) = I/I^2 \longrightarrow \Omega_{P/A} \otimes_P B$$

as a complex of B-modules with I/I^2 placed in degree -1. Since the term in degree 0 is free, this complex has tor-amplitude in $[-1, 0]$ if and only if I/I^2 is a flat B-module, see Lemma 0653. If this holds, then $NL(\alpha) \otimes^{\mathbf{L}}_B B' = NL(\alpha) \otimes_B B'$ and we get the second statement. □